

SOK3022

Topic 1: Time Series Analysis – VECM

An application to the relationships between CO2 emissions, financial development (FD), and GDP

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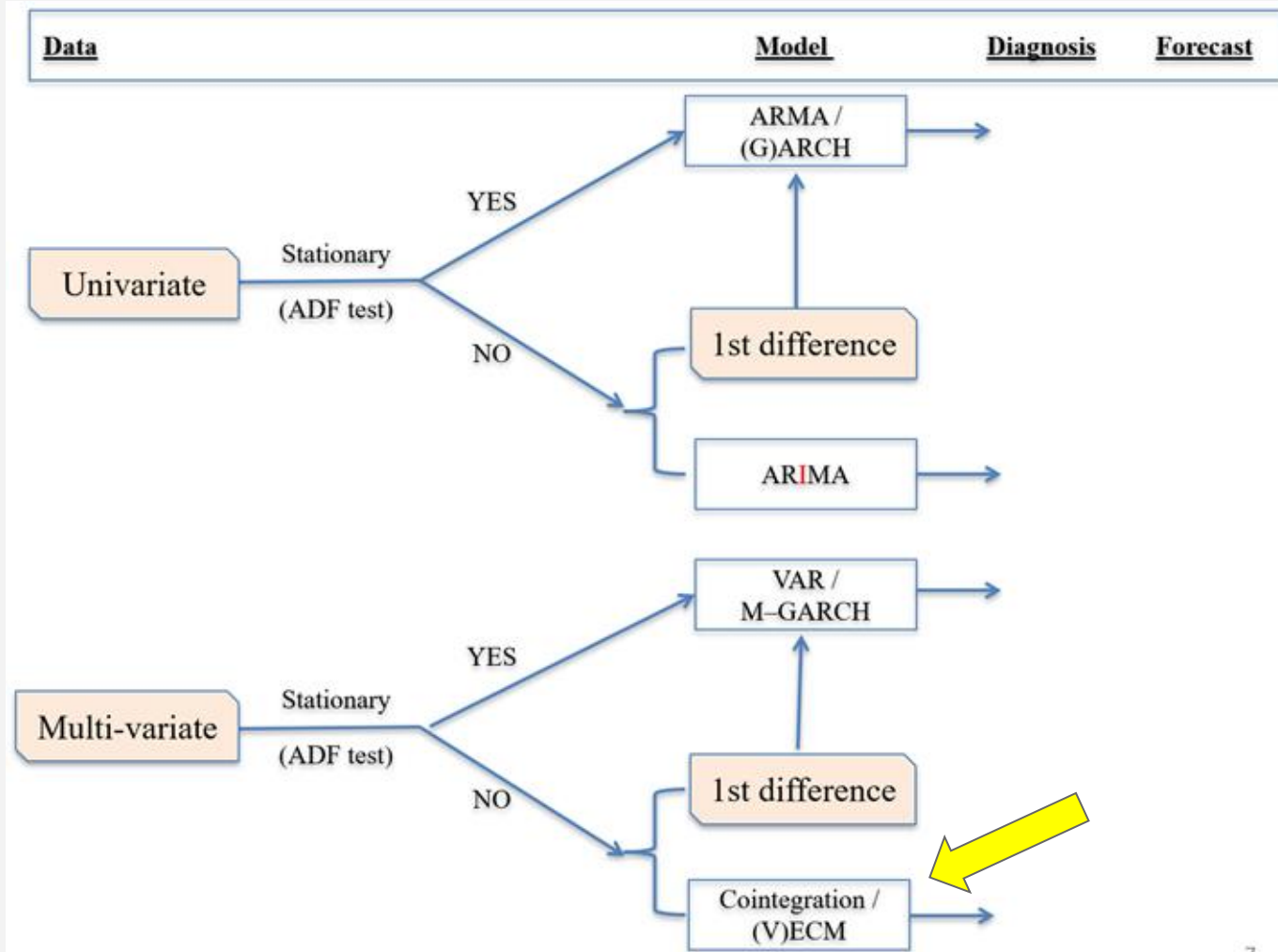
dengun.zhang@uis.no (preferred)

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- Part 1: Intro - Stationary, cointegration, and VECM (02.03)
- Part 2: Research topics and theories (03.03)
- Part 3: Previous studies (03.03)
- Part 4: Data and key variables (03.03)
- Part 5: Estimation and interpretation – Examples (03.03)
- Part 6: Preliminary results and presentation (04.03, Mandatory attendance and participation)

Quiz / Discussion

- For the ADF test, what is the null hypothesis?
- For a system with three non-stationary variables, is it possible to have three cointegrating relationships?
- For VECM (**v**ector **e**rror **c**orrection **m**odel), what does ‘error’ mean?



Part 1: Intro - VECM

Types of time series data

$$\{x_1, x_2, \dots, x_T\} \text{ or } \{x_t\}, t = 1, 2, \dots, T$$

- Stationary vs. Non-stationary time series

Stationary series

- A stationary series converges to its steady state. Stationarity is a weaker condition than normality and does not ensure the absence of autocorrelated residuals.
- A time series is said to be stationary if its statistical properties do not change over time.

Covariance Stationary Series

- Mean is time-invariant
- Variance is constant
- All covariances are constant
 - all autocorrelations are constant

Stationary series

A stochastic process having a finite mean and variance is **covariance stationary** if for all t and $t-s$,

$$1. E(y_t) = E(y_{t-s}) = \mu$$

$$2. E[(y_t - \mu)^2] = E[(y_{t-s} - \mu)^2]$$

$$\text{or } [\text{var}(y_t) = \text{var}(y_{t-s}) =]$$

$$3. E[(y_t - \mu)(y_{t-s} - \mu)] = E[(y_{t-j} - \mu)(y_{t-j-s} - \mu)] = \gamma_s$$

$$\text{or } \text{cov}(y_t, y_{t-s}) = \text{cov}(y_{t-j}, y_{t-j-s}) = \gamma_s$$

where μ , and γ_s are all constants.

Stationary Time Series

- $z_t = b_1 y_{t-1} + b_2 y_{t-2} + \dots + b_p y_{t-p} + e_t$
- Autocovariance: $\gamma_k = \text{Cov}[y_t, y_{t-k}]$
- Autocorrelation: $\rho_k = \gamma_k / \gamma_0$
- Stationary series: γ_k depends only on k , not on t

Non-Stationary

- A **non-stationary** time series is one whose statistical properties (such as mean, variance, and autocovariance) **change over time**.
 - **Trend Non-Stationarity**: The series has a deterministic trend (e.g., a linear or exponential trend over time).
 - **Unit Root Non-Stationarity**: The series follows a stochastic trend (random walk behavior).
 - **Structural Break Non-Stationarity**: The series changes behavior due to external shocks.
- A **unit root** is a specific type of non-stationarity caused by a stochastic (random) trend.

$$Y_t = Y_{t-1} + \epsilon_t$$

Unit root tests

- ADF test (Augmented Dickey–Fuller)
 - Null hypothesis: Non-stationarity
 - Fail to reject the null hypothesis ($|\text{test statistic}| < |\text{critical value}|$, $p\text{-value} > 0.05$): the time series is non-stationary.
 - Reject the null hypothesis ($|\text{test statistic}| > |\text{critical value}|$): the time series is stationary.
- PP test (Phillips-Perron)
 - Null hypothesis: Stationarity
 - Fail to reject the null hypothesis ($|\text{test statistic}| < |\text{critical value}|$): the time series contains is stationary.
 - Reject the null hypothesis ($|\text{test statistic}| > |\text{critical value}|$): the time series is non-stationary.

Augmented Dicky-Fuller test, $\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{s=1}^m a_s \Delta y_{t-s} + v_t$ for $H_0 : \gamma = 0$, $H_1 : \gamma < 0$. Reject the null if $\tau \leq \tau^c$. If H_0 is not rejected, it is non-stationary. Or as in this particular instance with the test `tseries::adf.test`, for low values of ρ , H_0 is not rejected.

White
noise

White noise (WN)

Extension

The building block for our time series models is the *white noise* process, which I'll denote ϵ_t . In the least general case,

$$\epsilon_t \sim \text{i.i.d. } N(0, \sigma_\epsilon^2)$$

Notice three implications of this assumption:

1. $E(\epsilon_t) = E(\epsilon_t \mid \epsilon_{t-1}, \epsilon_{t-2}, \dots) = E(\epsilon_t \mid \text{all information at } t-1) = 0$.
2. $E(\epsilon_t \epsilon_{t-j}) = \text{cov}(\epsilon_t \epsilon_{t-j}) = 0$
3. $\text{var}(\epsilon_t) = \text{var}(\epsilon_t \mid \epsilon_{t-1}, \epsilon_{t-2}, \dots) = \text{var}(\epsilon_t \mid \text{all information at } t-1) = \sigma_\epsilon^2$

The first and second properties are the absence of any *serial correlation* or *predictability*. The third property is *conditional homoskedasticity* or a constant conditional variance.

Cointegration: basic concept

1. Cointegration refers to a **linear** combination of non-stationary variables.

- If $(\beta_1, \beta_2, \dots, \beta_n)$ is a cointegrating vector, then for any non-zero value of λ , $(\lambda\beta_1, \lambda\beta_2, \dots, \lambda\beta_n)$ is also a cointegrating vector.
- Typically, one of the variables is used to *normalize* the cointegrating vector by fixing its coefficient at unity.
 - To normalize the cointegrating vector with respect to x_{1t} , simply select $\lambda = 1/\beta_1$.

2. The equation must be balanced in that the order of integration of the two sides must be equal

3. If x_t has m components, there may be as many as $m-1$ linearly independent cointegrating vectors

Cointegration: Johansen test

For a n -variable first-order VAR given:

$x_t = A_1 x_{t-1} + \varepsilon_t$. Subtract x_{t-1} from each side to obtain:

$$\begin{aligned}\Delta x_t &= A_1 x_{t-1} - x_{t-1} + \varepsilon_t \\ &= (A_1 - I) x_{t-1} + \varepsilon_t \\ &= \pi x_{t-1} + \varepsilon_t\end{aligned}$$

The rank of $(A_1 - I)$ equals the number of cointegrating vectors.

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$$

$$\lambda_{\text{max}}(r, r+1) = -T \ln(1 - \hat{\lambda}_{r+1})$$

The trace test and the max-eigenvalue cointegration rank test.

Equilibrium error

- Letting β and x_t denote the vectors $(\beta_1, \beta_2, \dots, \beta_n)$ and $(x_{1t}, x_{2t}, \dots, x_{nt})$, the system is in long-run equilibrium when $\beta x_t' = 0$.
- The deviation from long-run equilibrium--called the **equilibrium error**.
- If the equilibrium is meaningful, it must be the case that the equilibrium error process is stationary.

VECM: vector error correction model

A VEC model is used when the variables in a system are cointegrated. The general form of a VEC model for two variables (X_t and Y_t) is:

$$\Delta Y_t = \alpha_y(Y_{t-1} - \beta X_{t-1}) + \sum_{i=1}^{p-1} \gamma_{yi} \Delta Y_{t-i} + \sum_{i=1}^{p-1} \delta_{xi} \Delta X_{t-i} + \epsilon_{yt}$$

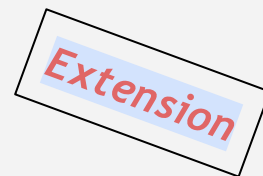
$$\Delta X_t = \alpha_x(Y_{t-1} - \beta X_{t-1}) + \sum_{i=1}^{p-1} \gamma_{xi} \Delta Y_{t-i} + \sum_{i=1}^{p-1} \delta_{xi} \Delta X_{t-i} + \epsilon_{xt}$$

Where:

- ΔY_t and ΔX_t : First differences of Y_t and X_t ,
- α_y and α_x : Adjustment coefficients to the long-term equilibrium,
- β : Cointegration parameter.

ECT
(error
correction
term)

Weak exogeneity and Granger causality



- A variable is **weakly exogenous** if it does not respond to the error correction term.
 - Null Hypothesis: Adjustment coefficient is zero ($\alpha=0$)
- Granger causality in a VEC model examines whether lagged values of one variable and/or the error correction term (ECT) can predict another variable.
 - Null Hypothesis: Lagged differences have no effect ($\delta=0$)

Zhang, D., Myrland, O., & Xie, J. (2016). Firm size, commodity price, and interdependence between firm-level stock prices: The case of Norwegian salmon industry. *Applied Economics and Finance*, 3(4), 179-189.

Abstract

This study aims to investigate the interdependence between stock prices of salmon firms listed at Oslo Stock Exchange, using cointegration techniques. The time path of the relationship between those stock prices is further examined by recursive estimations. During the sample period, we fail to confirm a common stochastic trend between the salmon firm prices with the exception of several time points. This is probably due to the differences between salmon stock firms' responses to commodity market price. It might also relate to the pairwise (bivariate) cointegration space within the multivariate system.

Key words: stock price; commodity price; salmon; interdependence; cointegration

Introduction

Salmon industry and prices

Econometric Methods

✓ Empirical Findings

Integration analysis of individual series

Cointegration analysis for stock prices

The impact of MHG on other stock prices

The impact of salmon market price on salmon stock prices

Conclusion

References

Table 1. Descriptive statistics of price variables

Table 2. Unit root tests

Table 3. Johansen test statistics for the multivariate system of stock prices

Table 4. Bivariate Johansen test statistics (salmon stock prices versus MHG)

Table 5. Long-run impact of MHG price on other salmon stock prices

Table 6. Bivariate Johansen test statistics (salmon stock prices versus salmon market price)

Table 7. Long-run impact of salmon market price on salmon stock price

Figure 1. Weekly salmon stock price and salmon price: 2007–2013 (mean of week 25–28 of 2007 = 100).

Figure 2. Normalized trace test statistic values (5% significant level) from the recursive estimations.

Example

Misund, B., Molnár, P., Nguyen, Q. M., & Totland, E. (2025). Impact of rent taxation on Norwegian salmon farming companies. *Aquaculture Economics & Management*, 29(1), 1-20.

Table 1. Descriptive Statistics of Price Variables

Variable	Mean	Maximum	Minimum	Standard Deviation
P_{MHG} (Marine Harvest)	41.94	72.14	10.08	14.98
P_{LSG} (Lerøy Seafood)	112.27	193.25	41.23	33.56
P_{SALM} (SalMar)	42.27	67.50	22.20	10.58
P_{GSF} (Grieg Seafood)	12.34	24.00	3.06	5.37
P_{NOS} (Salmon commodity price)	29.85	33.60	18.17	6.31

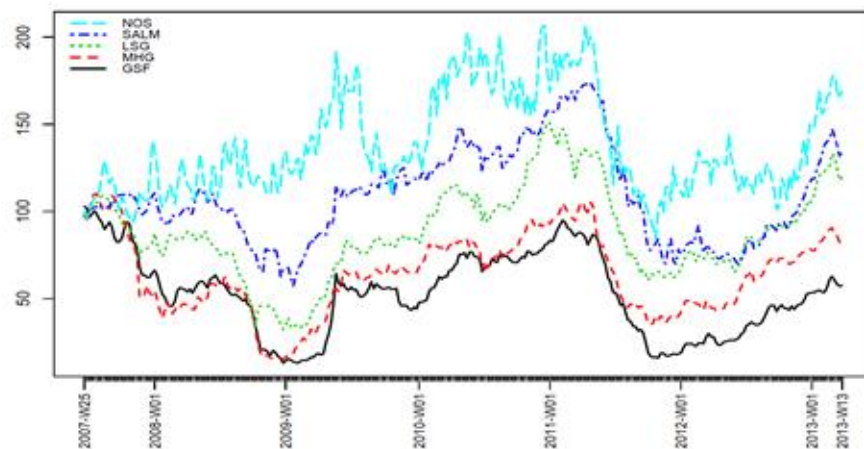


Figure 1. Weekly Salmon Stock Price and Salmon Price: 2007-2013 (week 25-28 of 2007 = 100).

Table 2. Unit Root Tests

Variable	ADF		KPSS	
	Level	Difference	Level	Difference
$\ln P_{MHG}$	-2.73	-5.25**	0.28**	0.089
$\ln P_{LSG}$	-1.85	-6.08**	0.33**	0.11
$\ln P_{SALM}$	-1.41	-13.40**	0.48**	0.12
$\ln P_{GSF}$	-2.34	-6.76**	0.29**	0.097
$\ln P_{NOS}$	-2.22	-16.93**	0.65**	0.055
Critical Value (5%)	-2.87	-2.87	0.146	0.146

Note: (i) the order of lagged difference terms with a maximum order of 6; (ii) ** indicates significance at the 5% level.

Table 6. Bivariate Johansen Test Statistics (Salmon Stock Prices Versus Salmon Commodity Price)

Trace Test			Maximum Eigenvalue Test		
H ₀ :	H _a :	Trace Statistic	H ₀ :	H _a :	Max-Eigen Statistic
$\ln P_{MHG}$ vs. $\ln P_{NOS}$					
$r = 0$	$r = 1$	14.41	$r = 0$	$r \geq 1$	9.32
$r = 1$	$r = 2$	5.09	$r \leq 1$	$r \geq 2$	5.09
$\ln P_{LSG}$ vs. $\ln P_{NOS}$					
$r = 0$	$r = 1$	13.86	$r = 0$	$r \geq 1$	7.26
$r = 1$	$r = 2$	6.59	$r \leq 1$	$r \geq 2$	6.59
$\ln P_{SALM}$ vs. $\ln P_{NOS}$					
$r = 0$	$r = 1$	293.37**	$r = 0$	$r \geq 1$	288.26**
$r = 1$	$r = 2$	5.10	$r \leq 1$	$r \geq 2$	5.10
$\ln P_{GSF}$ vs. $\ln P_{NOS}$					
$r = 0$	$r = 1$	42.25**	$r = 0$	$r \geq 1$	36.97**
$r = 1$	$r = 2$	5.28	$r \leq 1$	$r \geq 2$	5.28

Note: ** indicate significance at the 5% level.

Table 7. Long-Run Impact of Salmon Commodity Price on Salmon Stock Price

Variable	Coef.	Variable	Coef.
$\ln P_{SALM}$	1	$\ln P_{GSF}$	1
$\ln P_{NOS}$	-0.95*** (0.065)	$\ln P_{NOS}$	-1.46*** (0.39)
Trend	0.00040*** (0.00015)	Trend	0.0028** (0.00089)
Constant	-0.58	Constant	1.79

Note: ** and *** indicate significance at the 5% and 1% levels, respectively.

$$\begin{bmatrix} \Delta \ln P_{GSF,t} \\ \Delta \ln P_{NOS,t} \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} \beta_1 & \beta_2 \end{bmatrix} \begin{bmatrix} \ln P_{GSF,t-1} \\ \ln P_{NOS,t-1} \end{bmatrix} + \begin{bmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{bmatrix} \begin{bmatrix} \Delta \ln P_{GSF,t-1} \\ \Delta \ln P_{NOS,t-1} \end{bmatrix} + \begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix}$$

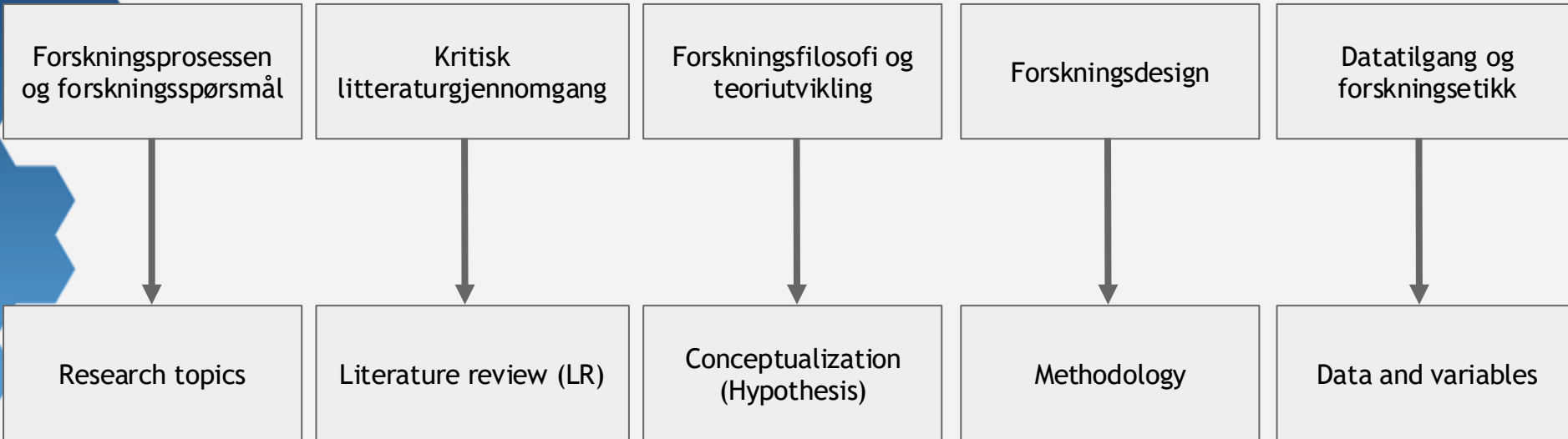
- Write down the long run (cointegrating) relationship, one equation.
- Write down the VECM for the two variables, two equations.

Why not use OLS for non-stationary data

- **Spurious regression.**
- **Violations of Classical Assumptions**
 - **Homoscedasticity Assumption (Constant Variance of Errors):** Non-stationary data often exhibit changing variances (heteroskedasticity)
 - **No Autocorrelation of Errors:** Non-stationary series typically exhibit **serial correlation**, leading to biased standard errors and invalid statistical inference.
 - **Normality of Residuals:** The residuals from an OLS regression with non-stationary variables are often non-normally distributed, making hypothesis tests invalid.

Part 2: Research topics and theories

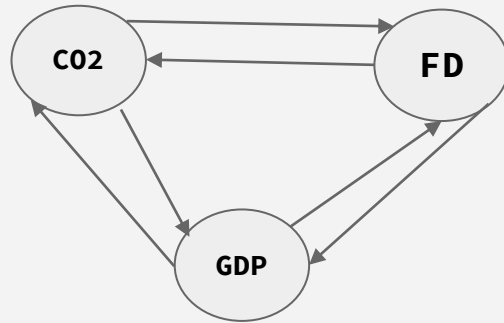
The Process of Research



Research objectives

For CO2 emissions (CO2), Financial Development (FD), and economic activity (GDP): (you can choose one of them for your presentation)

1. The relationship between CO2, FD, and GDP
 2. The relationship between CO2 and FD
 3. The relationship between CO2 and GDP
 4. The relationship between FD and GDP
- Developing countries vs. developed countries?



- CO2: CO2 emissions, FD: Financial development, GDP

Li, F., Wu, Y. C., Wang, M. C., Wong, W. K., & Xing, Z. (2021). Empirical study on CO2 emissions, financial development and economic growth of the BRICS countries. *Energies*, 14(21), 7341.

Sadorsky, P. (2010). The impact of financial development on energy consumption in emerging economies. *Energy Policy*, 38(5), 2528-2535.

Tao, M., Sheng, M. S., & Wen, L. (2023). How does financial development influence carbon emission intensity in the OECD countries: some insights from the information and communication technology perspective. *Journal of Environmental Management*, 335, 117553.

Zaidi, S. A. H., Zafar, M. W., Shahbaz, M., & Hou, F. (2019). Dynamic linkages between globalization, financial development and carbon emissions: evidence from Asia Pacific Economic Cooperation countries. *Journal of cleaner production*, 228, 533-543.

Kagochi, J. M., Nasser, O. M. A., & Kebede, E. (2013). Does financial development hold the key to economic growth? The case of Sub-Saharan Africa. *The Journal of Developing Areas*, 61-79.

CO2 vs. FD in general

- Financial Development (FD) influences resource allocation and investment in green technologies.
- Financial development can improve technological innovation and reduce energy consumption and thereby CO2 emissions.

However,

- Financial development makes it easier for companies to raise funds and expand production and
- for consumers to buy household appliances, increasing CO2 emissions (Sadorsky, 2010; Li et al., 2021).
- Developing countries vs. developed countries?

Sustainable finance





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COP29 UN Climate Conference Agrees to Triple Finance to Developing Countries, Protecting Lives and Livelihoods

“Triple finance to developing countries, from the previous goal of USD 100 billion annually, to USD 300 billion annually by 2035.

Secure efforts of all actors to work together to scale up finance to developing countries, from public and private sources, to the amount of USD 1.3 trillion per year by 2035.”

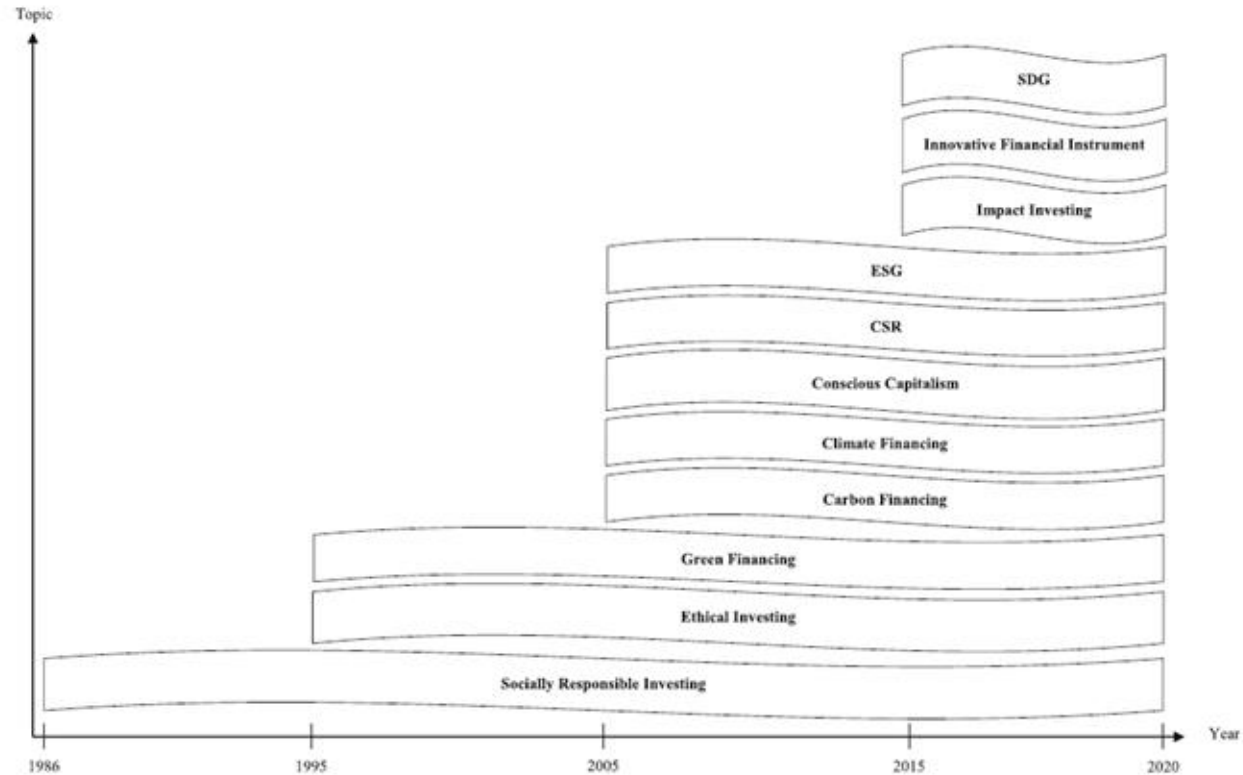


Fig. 1 Evolution of sustainable finance research. *CSR* Corporate social responsibility. *ESG* Environmental, social, and governance. *SDG* Sustainable development goals

CO2 vs. FD: OECD countries (Tao et al., 2023)



CO2 vs. FD: BRIC (Li et al., 2021)

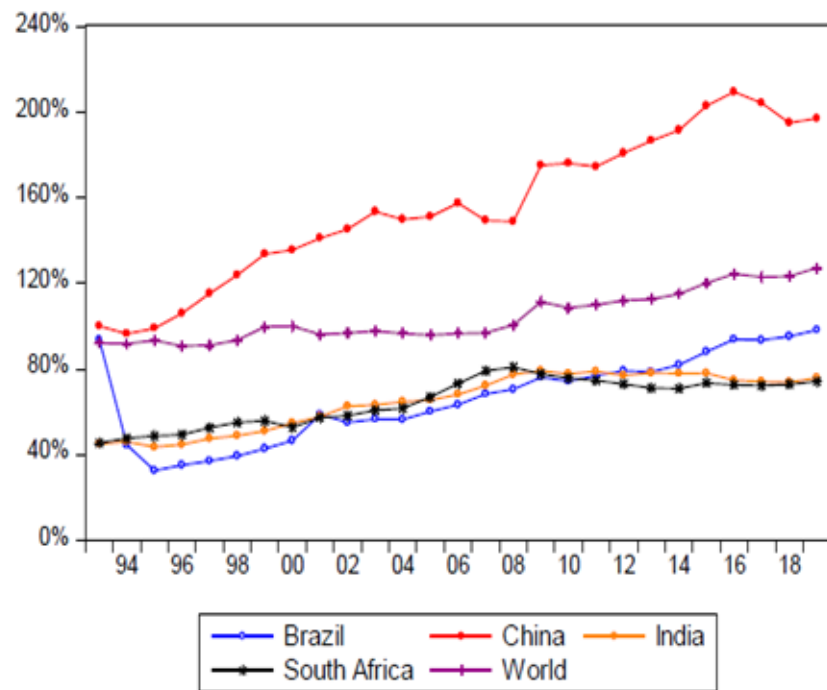


Figure 4. Broad money M2 (% of GDP).

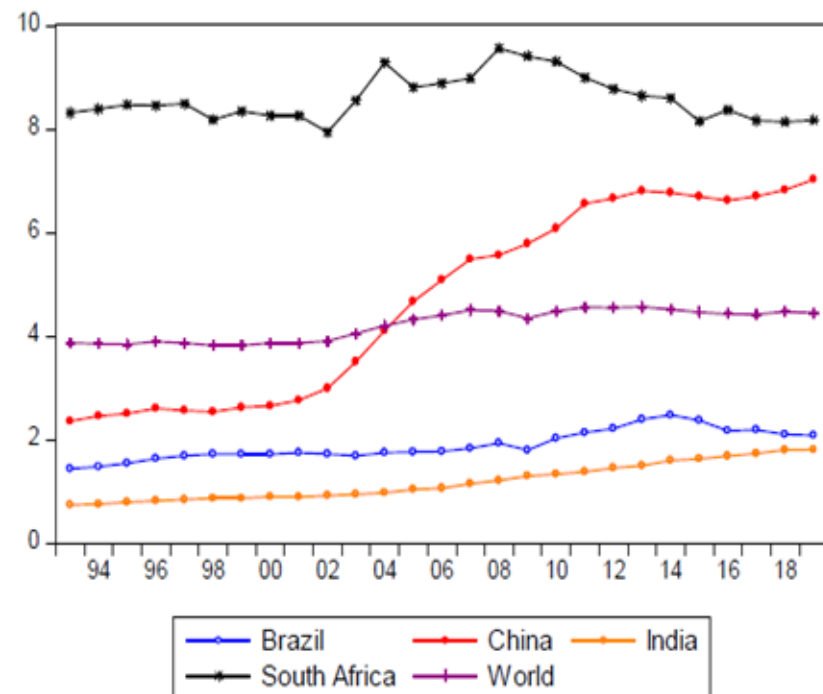


Figure 10. Per capita CO₂ emissions (tons per capita).

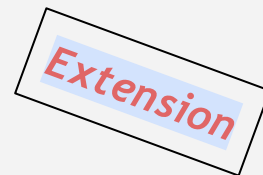
From FD to CO2

- Financial development leads to greater economic growth, leading to the positive impact of FD on energy demand and then emissions (Sadorsky, 2010):
 - For business
 - For consumers
- Financial development can improve technological innovation and reduce energy consumption and thereby CO2 emissions.

From CO₂ to FD

- CO₂ emissions and associated environmental risks can influence financial markets by altering investor sentiment and risk perceptions.
- Governmental regulations, such as carbon taxes or emissions trading systems (ETS), can redirect investments toward cleaner industries, influencing the structure and depth of financial markets.
- Foreign direct investment (FDI)

CO2 vs. FD at the firm level



- More generally, it is about how environmental performance (disclosures) affect access to finance.
- It is also related to the impact of corporate environmental performance on financial performance.

CO2 vs. FD at the firm level

- Environmental performance affects access to bank loans and contractual terms but varying effects in new EU members (Zhang, 2021)
- The impact of environmental performance on access to finance depends on the measures and financial sources (Zhang et al., 2022a).
- The role of environmental info disclosures on access to finance differs between firms led by females and males (Zhang et al., 2022b).
- The impact of environmental performance on financing choices differ between larger and small firms and firms in developing and developed countries (Zhang, 2024).

FD vs. GDP

- Financial markets play a vital role in the growth process by channeling funds to the most efficient investors and entrepreneurial innovation (Kagochi et al., 2013)
- Other factors: internal government policies, **political stability**, formation, development of human capital, banking infrastructure, export policies, and direct investment.

FD vs. GDP

- Two channels through which FD affects GDP (Sadorsky, 2010):
 - **Factor productivity:** lessen information asymmetries, learning to better monitoring and selecting investment projects.
 - **Factor accumulation:** Organized financial systems increase efficiency of previously unproductive resources (e.g., private finance).
- Economic growth leads to financial development: demand for financial services grows due to economic growth.

TABLE 2. DEFINITION OF VARIABLES

Variable	Definition	Source
Economic Growth (Y)	Growth in Per capita real GDP	WDI
Stock Market Capitalization (SMC)	Total market value of all listed shares / GDP	FSEDD
Stock Market Turnover Ratio (TR)	Value of trades of shares over market capitalization	FSEDD
Stock Market Value Traded (VT)	Value of trades of domestic stocks / GDP	FSEDD
Private Credit (PC)	Private Credit by Deposit Money Banks and Other Financial Institutions / GDP	FSEDD
Liquid Liabilities (LIQ)	Liquid Liabilities / GDP	FSEDD
Bank Assets (BA)	Deposit Money Bank Assets / GDP	FSEDD
School Enrollment (SCH)	Average years of Schooling	B&L
Capital Formation (K)	Net additions of capital stock that are used to provide goods and services.	UN
Natural growth (N)	The births and deaths in a country's population	WDI
Life Expectancy (L)	The average number of years a newborn infant would be expected to live.	WDI

**TABLE 6. PANEL GRANGER CAUSALITY TESTS
(STOCK MARKET INDICATORS)**

Null Hypothesis	<i>F- Statistics</i>	<i>p-value</i>
SMC does not Granger cause Y	3.4157	0.0020**
Y does not Granger cause SMC	2.4463	0.0678*
VT does not Granger cause Y	4.2603	0.0064***
Y does not Granger cause VT	2.4966	0.0077***
TR does not Granger cause Y	4.8712	0.0032***
Y does not Granger cause TR	3.4157	0.0026**

*Notes: Dynamics asterisks indicate rejection of the null hypotheses that there is no causal relationship between the two variables. ***Significant at the 1% level.*

CO2 vs. GDP: Environment Kuznets Curve (EKC)

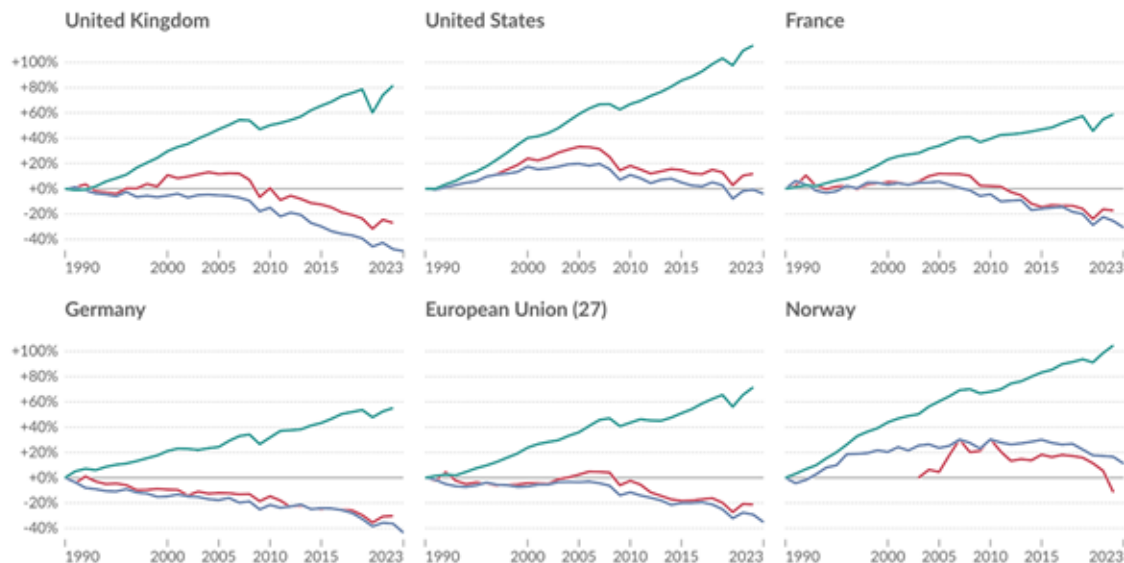
- GDP growth often leads to higher energy consumption and emissions; however,
- The EKC hypothesizes that as a country's economy grows, environmental degradation (such as CO2 emission) initially increases, but after a certain level of development, it begins to decrease.
- This means an inverted U-shaped curve for the relationship between GDP and CO2 emissions.



Change in CO₂ emissions and GDP

Consumption-based emissions¹ are national emissions that have been adjusted for trade. This measures fossil fuel and industry emissions². Land-use change is not included.

■ CO₂ emissions ■ Consumption-based CO₂ emissions ■ GDP



Data source: World Bank (2023); Global Carbon Budget (2024)

OurWorldInData.org/co2-and-greenhouse-gas-emissions | CC BY

Note: Gross Domestic Product (GDP) figures are adjusted for inflation.

1. Consumption-based emissions: Consumption-based emissions are national or regional emissions that have been adjusted for trade. They are calculated as domestic (or 'production-based' emissions) emissions minus the emissions generated in the production of goods and services that are exported to other countries or regions, plus emissions from the production of goods and services that are imported. Consumption-based emissions = Production-based - Exported + Imported emissions

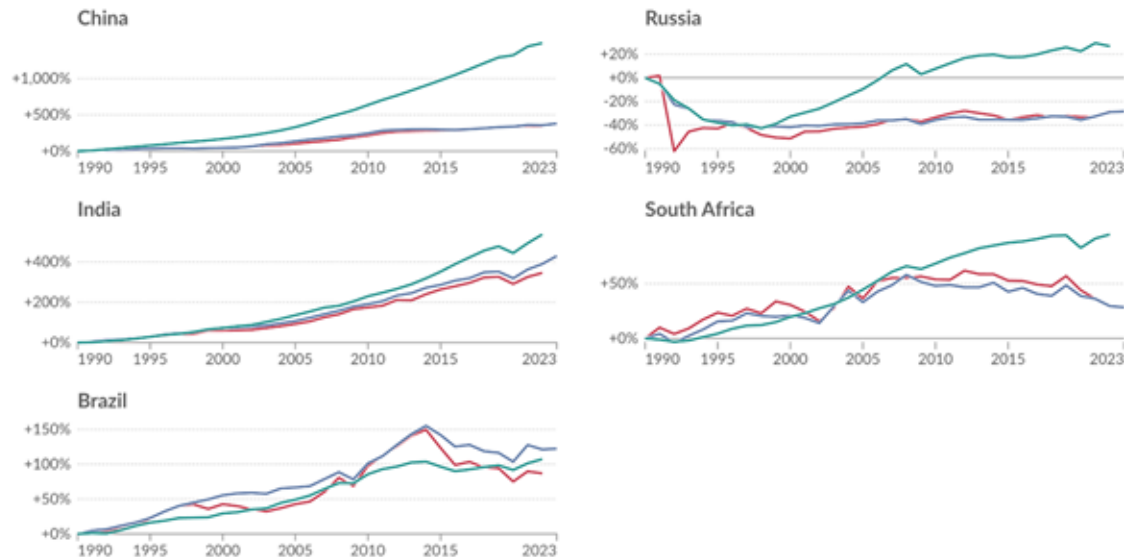
2. Fossil emissions: Fossil emissions measure the quantity of carbon dioxide (CO₂) emitted from the burning of fossil fuels, and directly from industrial processes such as cement and steel production. Fossil CO₂ includes emissions from coal, oil, gas, flaring, cement, steel, and other industrial processes. Fossil emissions do not include land use change, deforestation, soils, or vegetation.

Change in CO₂ emissions and GDP

Our World
in Data

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Tao et al. (2010): Environment Kuznets Curve

Example

M. Tao et al.

Journal of Environmental Management 335 (2023) 117553

Table 3
Benchmark regression.

	(1)	(2)	(3)	(4)	(5)	(6)
lnFDT_FD	-0.0642*** (0.015)	-0.0518*** (0.014)				
lnFDT_SZ			-0.0698*** (0.015)	-0.0547*** (0.014)		
lnFDT_EY					-0.0459** (0.023)	-0.0533** (0.024)
lnGI	-0.2833 (0.294)	0.8324*** (0.309)	-0.2592 (0.292)	0.8412*** (0.308)	-0.6414** (0.268)	-0.7589*** (0.277)
lnUR	2.0025*** (0.276)	2.6837*** (0.279)	2.0289*** (0.276)	2.7071*** (0.280)	0.8341*** (0.192)	1.9601*** (0.280)
lnPGDP	-1.8046*** (0.388)	-1.8324*** (0.358)	-1.8109*** (0.387)	-1.8486*** (0.358)	-2.2359*** (0.394)	-1.9611*** (0.392)
lnPGDP2	0.0632*** (0.019)	0.0742*** (0.018)	0.0633*** (0.019)	0.0749*** (0.018)	0.0866*** (0.019)	0.0711*** (0.019)
lnFCU	-0.0931 (0.076)	-0.0314 (0.067)	-0.0894 (0.076)	-0.0293 (0.067)	-0.1218* (0.074)	-0.1343* (0.077)
lnNE	5.3946*** (1.807)	5.4367*** (1.590)	5.5098*** (1.804)	5.5141*** (1.590)	3.5148* (1.845)	4.7013** (1.824)
_cons	5.5418* (2.994)	3.9329 (2.654)	5.4546* (2.986)	3.9304 (2.651)	10.3126*** (3.004)	7.1095** (3.011)
μ_i	✓	✓	✓	✓	✓	✓
λ_t	×	✓	×	✓	×	✓
Obs.	525	525	525	525	525	525
R-squared	0.6324	0.7274	0.6344	0.7280	0.6097	0.6224

Notes: Figures reported in () are standard errors; *p < 0.10, **p < 0.05, ***p < 0.01.

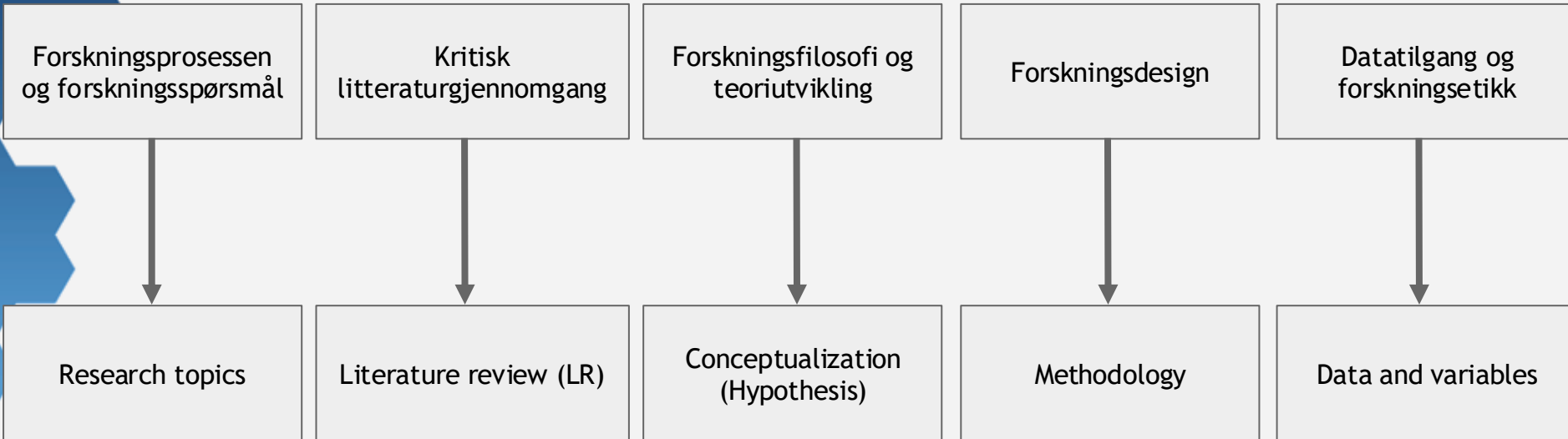
CO2 vs. GDP



- Causality direction: from CO2 emissions to GDP

Part 3: Previous studies on the relationship between CO₂ and FD

The Process of Research



Zaidi et al. (2019): Determinants of CO2

Zaidi, S. A. H., Zafar, M. W., Shahbaz, M., & Hou, F. (2019). Dynamic linkages between globalization, financial development and carbon emissions: evidence from Asia Pacific Economic Cooperation countries. *Journal of cleaner production*, 228, 533-543.

- ✓ Dynamic linkages between globalization, financial development and carbon emissions: Evidence from Asia Pacific Economic Coo ...
 - 1. Introduction
 - ✓ 2. Literature review
 - 2.1. Globalization–Carbon emissions nexus
 - 2.2. nexus of financial development & carbon emissions
 - 2.3. economic growth–carbon emissions nexus
 - ✓ 3. Data and theoretical framework
 - 3.1. Data
 - 3.2. Theoretical framework
 - ✓ 4. Results, discussion and policy implications
 - 4.1. long-run analysis
 - 4.2. panel granger causality analysis
 - 5. Conclusion
 - Declarations of interest
 - Acknowledgment
 - Appendix
 - References

Zaidi et al. (2019)

- Hypothesis: nexus of financial development & carbon emissions
Positive:
Negative:
- “3.2. Theoretical framework”
- Methods: Panel Granger causality analysis
- Data: APEC countries, panel data from 1990 to 2016.

Zaidi et al. (2019)

$$CO_2 = f(GLOB, FD, GDP, GDPS, EI) \quad (1)$$

where CO_2 is CO_2 emissions, GLOB is globalization, FD is financial development, GDP is gross domestic product, GDPS is squared term of GDP to measure EKC, and EI is energy intensity. To reduce the sharpness in the data, we take the natural-log of all the variables. In contrast with a simple linear transformation, log-linear model offers empirically consistent and efficient results.

Zaidi et al. (2019)

Table 4
Unit root tests results.

variables	CIPS		CADF	
	Level	First difference	Level	First difference
lnCO ₂	−1.937	−4.244 ***	−2.071	−3.170***
lnGLOB	−2.481	−5.150 ***	−2.316	−3.590***
lnFD	−1.954	−3.910 ***	−2.322	−3.041***
lnGDP	−2.178	−3.505 ***	−2.295	−2.921***
lnGDPS	−2.164	−3.454 ***	−2.382	−2.969***
lnEI	−1.877	−4.086 ***	−1.897	−3.129 ***

Note: Here CIPS of truncated test is used. *** show the rejection of null hypothesis. We included a Constant and Trend as suggested by [Pesaran \(2007\)](#). We reject the null hypothesis if data of atleast one country is stationary. Critical values of CIPS for 1% significance level is (−2.81) and for CADF (−2.58) for 1% significance level.

Zaidi et al. (2019)

Table 6

Long-run estimation results.

Variables	CUP-FM		CUP-BC		PMG	
	coefficient	t-statistics	coefficient	t-statistics	coefficient	t-statistics
lnGLOB	−0.0330**	2.5503	−0.0627***	3.0749	−0.086***	3.464
lnFD	−0.0021***	7.48577	−0.0089***	7.92114	−0.028**	−2.298
lnGDP	1.1209***	8.3359	1.1015***	8.4417	1.523***	4.899
lnGDPS	−0.0069***	15.4065	−0.0063***	15.8410	−0.029*	1.741
lnEI	0.9768***	24.1376	0.8901***	21.8802	0.968***	20.063

Note: ***, ** and * show significance at the 1%, 5% and 10% levels, respectively.

Tao et al. (2023)

- Tao, M., Sheng, M. S., & Wen, L. (2023). How does financial development influence carbon emission intensity in the OECD countries: some insights from the information and communication technology perspective. *Journal of Environmental Management*, 335, 117553.



How does financial development influence carbon emission intensity in the OECD countries: Some insights from the informatio ...

1 Introduction

> 2 Literature review

> 3 Methodology

> 4 Empirical findings

> 5 Mechanism analysis

6 Conclusions and policy implications

Credit author statement

Declaration of competing interest

Data availability

Acknowledgements

Appendix A Supplementary data

References

Tao et al. (2023)

- Hypothesis: 2.1. FDT-CEs nexus
Positive:
Negative:
- 2.4. Literature gaps
- Methods: Two-way fixed panel data model.
- Data: 35 OECD countries during 2004–2018

Tao et al. (2023)

Table 2
Descriptive statistics.

variables	definition	mean	sd	min	max	unit
CEI	Carbon intensity	0.372	0.221	0.063	1.624	%
FDT_FD	Financial deepening	1.024	0.457	0.002	3.046	%
FDT_SZ	Financial size	0.952	0.429	0.002	3.046	%
FDT_EY	Financial efficiency	0.950	0.250	0.334	2.141	%
GI	Government intervention	0.192	0.038	0.103	0.279	%
UR	Urbanisation rate	0.771	0.113	0.513	0.980	%
LNP GDP	Economic growth (logarithm)	10.451	0.406	9.304	11.670	US \$
FCU	Foreign capital utilisation	0.412	0.143	0.071	0.690	%
NE	Non-renewable energy consumption	0.059	0.052	0.0003	0.291	%
ITC1	Internet-based ICT	0.720	0.184	0.141	0.990	%
ICT2	Service-based ICT	0.395	0.170	0.006	0.796	%

Tao et al. (2023)

Table 3
Benchmark regression.

	(1)	(2)	(3)	(4)	(5)	(6)
lnFDT_FD	-0.0642*** (0.015)	-0.0518*** (0.014)				
lnFDT_SZ			-0.0698*** (0.015)	-0.0547*** (0.014)		
lnFDT_EY					-0.0459** (0.023)	-0.0533** (0.024)
lnGI	-0.2833 (0.294)	0.8324*** (0.309)	-0.2592 (0.292)	0.8412*** (0.308)	-0.6414** (0.268)	-0.7589*** (0.277)
lnUR	2.0025*** (0.276)	2.6837*** (0.279)	2.0289*** (0.276)	2.7071*** (0.280)	0.8341*** (0.192)	1.9601*** (0.280)
lnPGDP	-1.8046*** (0.388)	-1.8324*** (0.358)	-1.8109*** (0.387)	-1.8486*** (0.358)	-2.2359*** (0.394)	-1.9611*** (0.392)
lnPGDP2	0.0632*** (0.019)	0.0742*** (0.018)	0.0633*** (0.019)	0.0749*** (0.018)	0.0866*** (0.019)	0.0711*** (0.019)
lnFCU	-0.0931 (0.076)	-0.0314 (0.067)	-0.0894 (0.076)	-0.0293 (0.067)	-0.1218* (0.074)	-0.1343* (0.077)
lnNE	5.3946*** (1.807)	5.4367*** (1.590)	5.5098*** (1.804)	5.5141*** (1.590)	3.5148* (1.845)	4.7013** (1.824)
_cons	5.5418* (2.994)	3.9329 (2.654)	5.4546* (2.986)	3.9304 (2.651)	10.3126*** (3.004)	7.1095** (3.011)
μ_i	✓	✓	✓	✓	✓	✓
λ_t	×	✓	×	✓	×	✓
Obs.	525	525	525	525	525	525
R-squared	0.6324	0.7274	0.6344	0.7280	0.6097	0.6224

Notes: Figures reported in () are standard errors; *p < 0.10, **p < 0.05, ***p < 0.01.

Sadorsky (2010)



- Sadorsky, P. (2010). The impact of financial development on energy consumption in emerging economies. *Energy Policy*, 38(5), 2528-2535.
- What are the hypotheses?
- What kind of data are used in this article?
- Which econometrics are applied?
- The main findings?

Sadorsky (2010)

- What are the hypotheses?

demand. This study uses recently developed generalized method of moments estimation techniques that explicitly deal with endogeneity issues in dynamic models to examine the impact of financial development on energy consumption in a sample of emerging countries. In doing so, this paper takes an important step towards more fully understanding the relationship between financial development and energy demand.

2001). Consistent with this view that financial development leads to greater economic growth is the likelihood that energy demand should be positively impacted by increases in financial development. Increased financial development makes it easier for consumers and business to save, invest and borrow money. As

Another view is that economic growth leads to financial development (Robinson, 1952; Lucas, 1988; Stern, 1989). Financial development increases as the demand for financial services grows and the demand for financial services is directly tied to economic growth. In this view financial development follows economic growth. Consistent with this view is the likelihood that energy demand should be relatively non-responsive to financial development.

Sadorsky (2010)

- What kind of data/variables are used in this article?

This study uses annual data on energy demand and financial development for 22 emerging countries. The panel data set is an unbalanced panel of 22 emerging countries followed over 17 years (1990–2006). The dimensions of the panel data set are

Energy consumption is measured by energy use in kg of oil equivalent per capita (energy). Real GDP per capita is measured in

Foreign direct investment is one of the financial development indicators used in this paper. Foreign direct investment can

In addition to FDI, the following financial development variables are used in this paper, the ratio of deposit money bank assets to GDP (dbagdp) (Creane et al., 2007; Tamazian et al., 2009), the ratio of stock market capitalization to GDP (stmktcap),

the ratio of stock market total value traded to GDP (stvaltraded), and the stock market turnover ratio (stturnover). All of these financial development data are available from the Beck et al. (2000) Financial Structure Data base (updated November 21, 2008).

Sadorsky (2010)

- Which econometrics are applied?

The empirical model is specified as a reduced form dynamic panel model of energy demand. Energy demand (e) depends upon income (y), price (p), and a measure of financial development (d)

$$e_{it} = \alpha_i e_{it-1} + \beta_{i1} y_{it} + \beta_{i2} p_{it} + \beta_{i3} d_{it} + v_i + \psi_t + \varepsilon_{it} \quad (1)$$

where i denotes the country ($i=1, \dots, 22$) and t denotes the time period ($t=1990, \dots, 2006$). Eq. (1) is a fairly general specification which allows for dynamic energy demand effects, individual fixed country effects (v), fixed time effects (ψ), and a stochastic error term (ε).

Sadorsky (2010)

- The main findings?

Table 4

System GMM panel estimation regression results treating income as endogenous.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
lenergy(-1)	0.9581*** (46.56)	0.9578*** (46.98)	0.9496*** (43.52)	0.9763*** (57.94)	0.9702*** (79.17)	0.9543*** (61.06)
lgdppcap	0.0602** (2.20)	0.0605** (2.22)	0.0846*** (2.67)	0.0361 (1.32)	0.0364* (1.68)	0.0521*** (2.58)
lcpi	0.0135** (2.03)	0.0134** (2.03)	0.0188** (2.17)	0.0088* (1.65)	0.0072 (1.46)	0.0098 (1.51)
fdigdp		0.0004 (0.27)				
ldbaggdp			-0.0058 (-0.42)			
lstmkcap				0.0088** (2.04)		
lstvaltraded					0.0101*** (3.41)	
lstturnover						0.0121** (2.25)

Table 5

System GMM panel estimation regression results treating income and financial development as endogenous.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
lenergy(-1)	0.9581*** (46.56)	0.9635*** (52.01)	0.9565*** (65.74)	0.9847*** (66.23)	0.9731*** (89.67)	0.9588*** (71.73)
lgdppcap	0.0602** (2.20)	0.0447** (2.21)	0.0595*** (2.92)	0.0297 (1.17)	0.0245 (1.42)	0.0423*** (2.72)
lcpi	0.0135** (2.03)	0.0124** (2.1)	0.0137** (2.00)	0.0085 (1.61)	0.0074* (1.67)	0.0091 (1.41)
fdigdp		0.0007(0.37)				
ldbaggdp			0.0084(0.72)			
lstmkcap				0.0111*** (2.65)		
lstvaltraded					0.0129*** (5.43)	
lstturnover						0.0156*** (3.53)
AR(1)	-3.28 (0.00)	-3.29 (0.00)	-3.20 (0.00)	-3.30(0.00)	-3.28(0.00)	-3.26(0.00)
AR(2)	0.94 (0.35)	0.93 (0.35)	1.03 (0.30)	0.92 (0.36)	0.89 (0.38)	0.74 (0.46)
Sargan	135.23 (0.06)	178.31 (0.26)	169.17 (0.44)	190.18 (0.11)	162.25 (0.59)	165.85 (0.51)

Part 4: Data and key variables

Data sources

Data are at country level and by year.

WDI

<https://databank.worldbank.org/source/world-development-indicators>

<https://ourworldindata.org/>

<https://climatecommunication.yale.edu/>

Sample countries: BRICS (Brazil, Russia, India, China, and South Africa) vs. developed countries (OECD), such as the US, the UK, Germany, France, Japan, and Norway.

Key variables

- CO2 emissions (Alternative: energy consumption, energy intensity...)
- Financial development
- GDP (other economic activities: exports or imports)

Key variables from WDI

<https://databank.worldbank.org/source/world-development-indicators>

- CO2 emissions (CO2):
'EN.GHG.CO2.MT.CE.AR5': Carbon dioxide (CO2) emissions (total) excluding LULUCF (Mt CO2e)
- Financial development (FD)
'(FD)NE.GDI.TOTL.KD': Gross capital formation (constant 2015 US\$)
- GDP
'NY.GDP.MKTP.KD': GDP (constant 2015 US\$)



This page is in [English](#) [Español](#) [Français](#) [عربي](#) [中文](#)

DataBank World Development Indicators ⓘ

Table

Chart

9 M

Variables Layout Styles Save Share Embed

Database	Available	86	Selected	1
----------	-----------	----	----------	---

Country	Available	266	Selected	0
---------	-----------	-----	----------	---

All Countries Aggregates

Enter Keywords for  

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

☐ Afghanistan	☐ Albania
☐ Algeria	☐ American Samoa
☐ Andorra	☐ Angola
☐ Antigua and Barbuda	☐ Argentina
☐ Armenia	☐ Aruba
☐ Australia	☐ Austria
☐ Azerbaijan	☐ Bahamas, The
☐ Bahrain	☐ Bangladesh
☐ Barbados	☐ Belarus

Create Custom Group

Series	Available	Selected	0
--------	-----------	----------	---

Time Available Selected 0

Preview

Clear Selection | Add Country (0) Add Series (0) Add Time (0)

Please select variables from each of the following dimensions to view a report. You can select from left panel or by clicking the links above.

- Country
- Series
- Time

Apply Changes

Install/Load necessary libraries

```
library(tseries) # For ADF test
library(urca)     # For Johansen test
library(vars)     # For VECM wo SE
library(tsDyn)    # For VECM with SE for Alpha
library(readxl)   # Import xlsx data
library(car)      # Choose lag in VAR
library(aTSA)     # Engle-Granger(or EG) tests,
```

R-Code

```
#####
#### Import data and clean/organize data
#####

# D.CI <- read.csv("path_to_your_file/Data name.csv")
# or
# D.CI <- read_excel("path_to_your_file/Data name.xlsx", sheet='name of the sheet')
# D.CI = as.data.frame (D.CI); nrow(D.CI) # head(D.CI)

D.CI = read_excel("Data_World_Development_Indicators.xlsx",sheet='tilR');
D.CI = as.data.frame (D.CI); nrow(D.CI) # head(D.CI)

D.CI1 = D.CI[D.CI[,4]%in%c('EN.GHG.CO2.MT.CE.ARS', 'NE.GDI.TOTL.KD', 'NY.GDP.MKTP.KD'), ];
D.CI1[,1]=D.CI1[,3]=NULL

varialbe = as.factor(D.CI1[,2]); levels(varialbe)
levels(varialbe) = c('CO2', 'Fin', 'GDP')
D.CI1[,2] = varialbe

names(D.CI1) = c('Country', 'Series', c(1960:2023))# str(D.CI1)
#D.CI1 = remove.factors(D.CI1)

D.CI2 = reshape(D.CI1, direction = 'long', varying=list (names(D.CI1)[3:ncol(D.CI1)] ), v
D.CI3 = reshape(D.CI2, idvar=c("Country", "Year"), v.names=c('v'), timevar=c("Series"), di
unique(D.CI3$Country)
```

```
#####
```

```
#### Choose a sample country
```

```
#####
```

```
unique(D.CI3$Country)
```

```
Dat = D.CI3[D.CI3$Country=='USA',]; nrow(Dat)
```

```
Dat = Dat[Dat$v.CO2>0, ]; nrow(Dat)
```

```
Dat = Dat[Dat$v.GDP>0, ]; nrow(Dat)
```

```
Dat = Dat[Dat$v.Fin>0, ]; nrow(Dat)
```

```
Year = Dat$Year
```

```
Y = log(Dat$v.CO2)
```

```
X = log(Dat$v.Fin);
```

```
Z = log(Dat$v.GDP)
```

```
> D.CI = read_excel("Data_World_Development_Indicators.xlsx",sheet='tilR');
> D.CI = as.data.frame (D.CI); nrow(D.CI) # head(D.CI)
[1] 13464
>
> D.CI1 = D.CI[D.CI[,4]%in%c('EN.GHG.CO2.MT.CE.AR5', 'NE.GDI.TOTL.KD', 'NY.GDP.MKTP.KD')];
[1] 13464
> D.CI1[,1]=D.CI1[,3]=NULL
>
> varialbe = as.factor(D.CI1[,2]); levels(varialbe)
[1] "EN.GHG.CO2.MT.CE.AR5" "NE.GDI.TOTL.KD"      "NY.GDP.MKTP.KD"
> levels(varialbe) = c('CO2','Fin','GDP')
> D.CI1[,2] = varialbe
>
> names(D.CI1) = c('Country','Series',c(1960:2023))# str(D.CI1)
> #D.CI1 = remove.factors(D.CI1)
>
> D.CI2 = reshape(D.CI1, direction = 'long', varying=list (names(D.CI1)[3:ncol(D.CI1)]))
> D.CI3 = reshape(D.CI2, idvar=c("Country","Year"), v.names=c('v'), timevar=c("Series"))
> unique(D.CI1$Country)
[1] "CHN" "BRA" "IND" "ZAF" "RUS" "USA" "NOR" "FRA" "GBR"
>
```

```
#####
```

```
##### Choose a sample country
```

```
#####
```

```
unique(D.CI3$Country)
```

```
Dat = D.CI3[D.CI3$Country=='USA',]; nrow(Dat)
```

```
Dat = Dat[Dat$v.C02>0, ]; nrow(Dat)
```

```
Dat = Dat[Dat$v.GDP>0, ]; nrow(Dat)
```

```
Dat = Dat[Dat$v.Fin>0, ]; nrow(Dat)
```

```
Year = Dat$Year
```

```
C02 = log(Dat$v.C02); summary(C02)
```

```
FD = log(Dat$v.Fin); summary(FD)
```

```
GDP = log(Dat$v.GDP); summary(GDP)
```

Part 5: Estimation and interpretation – Examples

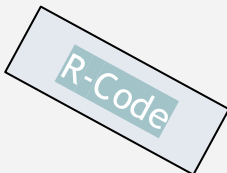
“For your replication”

USA vs. India

- Hypotheses:
 - Different impacts of FD (financial development) on CO2
 - Different impacts on of GDP on CO2
- Data
 - WTI:
 - Period:
- Measurement
 - CO2: CO2 emissions
 - FD: Financial development
 - GDP
- Methods
 - ADF – Cointegration test – VECM of three variables.

Account ownership at a financial institution or with a mobile-money-service provider, female (% of population ages 15+) ▼	FX.OWN.TOTL.FE.ZS ▼	▼
Adjusted savings: consumption of fixed capital (% of GNI)	NY.ADJ.DKAP.GN.ZS	Saving
Carbon dioxide (CO2) emissions excluding LULUCF per capita (t CO2e/capita)	EN.GHG.CO2.PC.CE.AR5	CO2
Carbon intensity of GDP (kg CO2e per constant 2015 US\$ of GDP)	EN.GHG.CO2.RT.GDP.KD	Intensity
Consumer price index (2010 = 100)	FP.CPI.TOTL	CPI
Domestic credit provided by financial sector (% of GDP)	FS.AST.DOMS.GD.ZS	Credit.Fin
Domestic credit to private sector (% of GDP)	FS.AST.PRVT.GD.ZS	Credit
Domestic credit to private sector by banks (% of GDP)	FD.AST.PRVT.GD.ZS	Credit.Bank
Energy use (kg of oil equivalent per capita)	EG.USE.PCAP.KG.OE	Energy
Exports of goods and services (% of GDP)	NE.EXP.GNFS.ZS	Export
Foreign direct investment, net inflows (% of GDP)	BX.KLT.DINV.WD.GD.ZS	FDI
GDP per capita (constant 2015 US\$)	NY.GDP.PCAP.KD	GDP
Gross capital formation (% of GDP)	NE.GDI.TOTL.ZS	Capital
Gross fixed capital formation (% of GDP)	NE.GDI.FTOT.ZS	Capital.Fixed
Gross fixed capital formation, private sector (% of GDP)	NE.GDI.FPRV.ZS	Capital.Fixed2
Market capitalization of listed domestic companies (% of GDP)	CM.MKT.LCAP.GD.ZS	Mcap
Monetary Sector credit to private sector (% GDP)	FM.AST.PRVT.GD.ZS	Monetary
Series Name	Series Code	Variable
Stocks traded, total value (% of GDP)	CM.MKT.TRAD.GD.ZS	Stock
Total greenhouse gas emissions excluding LULUCF per capita (t CO2e/capita)	EN.GHG.ALL.PC.CE.AR5	GHG

USA: ADF test



```
> Dat = D.CI3[D.CI3$Country=='USA',]; nrow(Dat)
[1] 64
> Dat = Dat[Dat$v.CO2>0, ]; nrow(Dat)
[1] 53
> Dat = Dat[Dat$v.GDP>0, ]; nrow(Dat)
[1] 53
> Dat = Dat[Dat$v.Fin>0, ]; nrow(Dat)
[1] 53
> Year = Dat$Year
> CO2 = log(Dat$v.CO2); summary(CO2)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
8.401   8.472   8.522   8.538   8.599   8.688
> FD = log(Dat$v.Fin); summary(FD)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
27.46   27.88   28.41   28.38   28.84   29.20
> GDP = log(Dat$v.GDP); summary(GDP)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
29.27   29.64   30.08   30.05   30.42   30.70
```

```
> adf.test(CO2)
```

Augmented Dickey-Fuller Test
alternative: stationary

Type 1: no drift no trend

	lag	ADF	p.value
[1,]	0	0.1980	0.697
[2,]	1	0.3141	0.730
[3,]	2	0.0729	0.661
[4,]	3	-0.1581	0.595

Type 2: with drift no trend

	lag	ADF	p.value
[1,]	0	-1.69	0.448
[2,]	1	-1.88	0.373
[3,]	2	-1.15	0.641
[4,]	3	-1.26	0.603

Type 3: with drift and trend

	lag	ADF	p.value
[1,]	0	-1.308	0.852
[2,]	1	-1.360	0.830
[3,]	2	-0.192	0.990
[4,]	3	-0.663	0.968

Note: in fact, p.value = 0.01 means p.value <= 0.01

```
> adf.test(FD)
```

Augmented Dickey-Fuller Test
alternative: stationary

USA: Johansen test

```
> J_test7 <- ca.jo(data.frame(CO2,FD,GDP), type = "eigen", e
```

```
#####  
# Johansen-Procedure #  
#####
```

Test type: maximal eigenvalue statistic (lambda max) , without

Eigenvalues (lambda):
[1] 6.976455e-01 2.593303e-01 7.285812e-02 2.853565e-17

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 2	3.86	7.52	9.24	12.97
r <= 1	15.31	13.75	15.67	20.20
r = 0	61.00	19.77	22.00	26.81

```
> J_test8 <- ca.jo(data.frame(CO2,FD,GDP), type = "trace", e
```

```
#####  
# Johansen-Procedure #  
#####
```

Test type: trace statistic , without linear trend and constant

Eigenvalues (lambda):
[1] 6.976455e-01 2.593303e-01 7.285812e-02 2.853565e-17

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 2	3.86	7.52	9.24	12.97
r <= 1	19.17	17.85	19.96	24.60
r = 0	80.17	32.00	34.91	41.07

Eigenvectors, normalised to first column:
(These are the cointegration relations)

USA: VECM

```
> Vecm2 = VECM(data.frame(CO2,FD,GDP), lag = 1, r=1, estim = "ML",include=c('const') )# ?VECM()
> summary(Vecm2)
```

```
#####
```

```
###Model VECM
```

```
#####
```

```
Full sample size: 53      End sample size: 51
```

```
Number of variables: 3  Number of estimated slope parameters 15
```

```
AIC -1162.56      BIC -1129.719      SSR 0.2773344
```

```
Cointegrating vector (estimated by ML):
```

```
      CO2      FD      GDP
r1    1 -9.645464 12.03064
```

	ECT	Intercept	CO2 -1	FD -1	GDP -1
Equation CO2	0.0129(0.0110)	-1.2665(1.0706)	-0.2467(0.2547)	-0.1954(0.1655)	1.2400(0.7637)
Equation FD	0.0606(0.0202)**	-5.8627(1.9613)**	-0.4659(0.4665)	-0.1157(0.3032)	2.3566(1.3991).
Equation GDP	0.0040(0.0064)	-0.3654(0.6235)	-0.0591(0.1483)	-0.0102(0.0964)	0.3362(0.4448)

USA: Long-run & Short-run

$$CO_2 - 0.96FD + 12.03GDP = 0$$

$$CO_2 = 0.96FD - 12.03GDP$$

$$\Delta CO_2 = -1.26 + 0.013ECT - 0.247\Delta CO_2_{t-1} - 0.195\Delta FD + 1.24\Delta GDP$$

$$\Delta FD = -5.86 + 0.061ECT - 0.47\Delta CO_2 - 0.116FD_{t-1} + 2.357\Delta GDP$$

$$\Delta GDP = -0.365 + 0.004ECT - 0.059\Delta CO_2 - 0.01FD + 0.336\Delta GDP_{t-1}$$

```
> round(Vecm2$model.specific$beta,3)
      r1
CO2  1.000
FD   -9.645
GDP  12.031
> cov = cov( residuals (Vecm2))
> t_stats <- Vecm2$model.specific$beta / sqrt(diag(cov))
> p_values <- 2 * (1 - pt(abs(t_stats), df = length(CO2) - 2))
> p_values
      r1
CO2  0
FD   0
GDP  0
>
> # $ short-run, alpha
>
> round ( summary(Vecm2)$coefMat, 3)
```

	Estimate	Std. Error	t value	Pr(> t)
CO2:ECT	0.013	0.011	1.169	0.248
CO2:Intercept	-1.267	1.071	-1.183	0.243
CO2:CO2-1	-0.247	0.255	-0.969	0.338
CO2:FD-1	-0.195	0.166	-1.180	0.244
CO2:GDP-1	1.240	0.764	1.624	0.111
FD:ECT	0.061	0.020	3.001	0.004
FD:Intercept	-5.863	1.961	-2.989	0.004
FD:CO2-1	-0.466	0.467	-0.999	0.323
FD:FD-1	-0.116	0.303	-0.382	0.704
FD:GDP-1	2.357	1.399	1.684	0.099
GDP:ECT	0.004	0.006	0.621	0.538
GDP:Intercept	-0.365	0.623	-0.586	0.561
GDP:CO2-1	-0.059	0.148	-0.399	0.692
GDP:FD-1	-0.010	0.096	-0.105	0.917
GDP:GDP-1	0.336	0.445	0.756	0.454

IND: ADF test

```
> Dat = D.CI3[D.CI3$Country=='IND',]; nrow(Dat)
[1] 64
> Dat = Dat[Dat$v.CO2>0, ]; nrow(Dat)
[1] 53
> Dat = Dat[Dat$v.GDP>0, ]; nrow(Dat)
[1] 53
> Dat = Dat[Dat$v.Fin>0, ]; nrow(Dat)
[1] 53
>
> Year = Dat$Year
> CO2 = log(Dat$v.CO2); summary(CO2)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
5.366   5.926   6.726   6.666   7.405   7.898
> FD = log(Dat$v.Fin); summary(FD)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
24.17   24.72   25.59   25.78   26.92   27.62
> GDP = log(Dat$v.GDP); summary(GDP)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
26.03   26.49   27.19   27.25   27.98   28.72
```

```
> # $ ADF Test for stationary
> #####
>
> adf.test(CO2)
Augmented Dickey-Fuller Test
alternative: stationary
```

Type 1: no drift no trend

	lag	ADF	p.value
[1,]	0	10.05	0.99
[2,]	1	5.00	0.99
[3,]	2	3.64	0.99
[4,]	3	2.99	0.99

Type 2: with drift no trend

	lag	ADF	p.value
[1,]	0	-0.599	0.834
[2,]	1	-0.901	0.728
[3,]	2	-0.941	0.714
[4,]	3	-1.438	0.539

Type 3: with drift and trend

	lag	ADF	p.value
[1,]	0	-1.38	0.823
[2,]	1	-1.47	0.785
[3,]	2	-1.77	0.663
[4,]	3	-1.37	0.827

Note: in fact, p.value = 0.01 means p.value <= 0.01

```
> adf.test(FD)
Augmented Dickey-Fuller Test
alternative: stationary
```

IND: Johansen test

```
>  
> J_test7 <- ca.jo(data.frame(CO2,FD,GDP), type = "eigen", ec
```

```
#####  
# Johansen-Procedure #  
#####
```

Test type: maximal eigenvalue statistic (lambda max) , without

Eigenvalues (lambda):

```
[1] 4.662112e-01 2.618799e-01 8.107173e-02 1.624830e-15
```

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 2	4.31	7.52	9.24	12.97
r <= 1	15.49	13.75	15.67	20.20
r = 0	32.02	19.77	22.00	26.81

```
> J_test8 <- ca.jo(data.frame(CO2,FD,GDP), type = "tr
```

```
#####  
# Johansen-Procedure #  
#####
```

Test type: trace statistic , without linear trend and

Eigenvalues (lambda):

```
[1] 4.662112e-01 2.618799e-01 8.107173e-02 1.624830e-
```

Values of teststatistic and critical values of test:

	test	10pct	5pct	1pct
r <= 2	4.31	7.52	9.24	12.97
r <= 1	19.80	17.85	19.96	24.60
r = 0	51.81	32.00	34.91	41.07

Eigenvectors, normalised to first column:
(These are the cointegration relations)

IND: VECM

```
> # $ VECM Function
> #####
>
> Vecm2 = VECM(data.frame(CO2,FD,GDP), lag = 1, r=1, estim = "ML",include=c('const'))# ?VECM()
> summary(Vecm2)
#####
###Model VECM
#####
Full sample size: 53    End sample size: 51
Number of variables: 3  Number of estimated slope parameters 15
AIC -964.5577    BIC -931.7167    SSR 0.4658934
Cointegrating vector (estimated by ML):
      CO2      FD      GDP
r1    1 -1.827023 1.81885

      ECT          Intercept      CO2 -1      FD -1      GDP -1
Equation CO2 -0.0223(0.0135)  0.2426(0.1225).  0.0236(0.1527)  -0.0019(0.0642)  0.1762(0.2047)
Equation FD  0.0653(0.0369).  -0.5219(0.3343)  0.2638(0.4168)  -0.0899(0.1753)  -0.2799(0.5587)
Equation GDP 0.0323(0.0117)** -0.2468(0.1059)*  0.3657(0.1320)**  0.0597(0.0555)  -0.3315(0.1770).
```

>

IND: Long-run vs. Short-run

$$CO2 - 1.827FD + 1.819GDP = 0$$

$$CO2 = 1.827FD - 1.819GDP$$

$$\Delta CO2 = 0.243 - 0.022ECT + 0.024\Delta CO2_{t-1} - 0.002\Delta FD + 0.176\Delta GDP$$

```
> # $ long-run, beta
>
> round(Vecm2$model.specific$beta,3)
      r1
CO2  1.000
FD   -1.827
GDP   1.819
> cov = cov( residuals (Vecm2))
> t_stats <- Vecm2$model.specific$beta / sqrt(diag(cov))
> p_values <- 2 * (1 - pt(abs(t_stats), df = length(CO2) - 2))
> p_values
      r1
CO2   0
FD     0
GDP   0
>
> # $ short-run, alpha
>
> round ( summary(Vecm2)$coefMat, 3)
              Estimate Std. Error t value Pr(>|t|)
CO2:ECT          -0.022      0.014  -1.647   0.106
CO2:Intercept     0.243      0.122   1.981   0.054
CO2:CO2-1         0.024      0.153   0.154   0.878
CO2:FD-1         -0.002      0.064  -0.030   0.976
CO2:GDP-1         0.176      0.205   0.861   0.394
FD:ECT           0.065      0.037   1.770   0.083
FD:Intercept    -0.522      0.334  -1.561   0.125
FD:CO2-1         0.264      0.417   0.633   0.530
FD:FD-1         -0.090      0.175  -0.513   0.611
FD:GDP-1        -0.280      0.559  -0.501   0.619
GDP:ECT          0.032      0.012   2.765   0.008
GDP:Intercept   -0.247      0.106  -2.331   0.024
GDP:CO2-1        0.366      0.132   2.770   0.008
GDP:FD-1         0.060      0.056   1.076   0.288
GDP:GDP-1       -0.331      0.177  -1.873   0.067
```

Main findings: USA vs. IND

- For a system of CO₂, FD (financial development), and GDP:
 - All variables are not stationary, $I(1)$, for both USA and IND
 - There is a cointegrating relationship among them, for both USA and IND.
- In the long run:
 - FD has a stronger positive impact on CO₂ in IND than in USA
- In the short run:
 - FD has a weaker and negative impact on CO₂ in IND than in USA, but both are insignificant.

Part 6: Preliminary results and presentation

Presentation

- Identify a hypothesis regarding the relationship between CO₂, FD, and GDP.
- Choose one OECD country and/or one non-OECD country.
- Choose data series for CO₂ emissions (or energy consumption), FD, and GDP (you can use the WDI data, as discussed in the lecture).
- Conduct ADF test for the three variables.
- Perform the Johansen test for the pairs of the three variables and a system with the three variables. (Totally, there are four tests. Ideally, at least one of the tests confirms a cointegration relationship)
- Estimate VECM for the system with the three variables or one of the pairwises.
- Interpret the results and conclude whether they support the hypothesis or not.

Presentation

- 10 minutes presentation
- 5 minutes for comments and suggestions.